

The Eisenstein-Ulam Spiral: Visualizing Prime Decomposition on Cyclotomic Lattices

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Interactive Visualization at:

<https://cyclotomic-ulam.vercel.app/>

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Abstract

This work introduces an interactive visualization of the Eisenstein–Ulam spiral, a hexagonal analogue of Ulam’s square spiral that reveals the distribution of Eisenstein primes and their relationship to the classical Fermat- and Gaussian-type prime classifications. By placing integers on a hexagonal lattice following Ulam’s spiral construction and colour-coding them according to their behaviour modulo 12, the visualization exposes striking filament patterns revealing that Eisenstein primes ($p \equiv 2 \pmod{3}$) form distinct hexagonal ray structures, while non-Eisenstein primes occupy complementary bands—a geometric manifestation of cubic reciprocity.

Introduction and Motivation

In 1963, Stanisław Ulam noticed striking diagonal structures when marking primes on a square spiral pattern [1, 2]. When integers are arranged in a square spiral and primes are marked, unexpected diagonal patterns emerge. This simple visualization revealed that primes, despite their apparent randomness, exhibit structured distribution patterns related to quadratic residues.

The Eisenstein-Ulam construction places integers on the lattice of $\mathbb{Z}[\omega]$, the ring of Eisenstein integers generated by a primitive cube root of unity [5, 7]. Thus the Eisenstein-Ulam spiral extends Ulam’s idea to the hexagonal lattice, which is the natural geometric setting for Eisenstein integers $\mathbb{Z}[\omega]$, where $\omega = e^{2\pi i/3} = \frac{-1+i\sqrt{3}}{2}$ is a primitive cube root of unity. This choice is motivated by several key mathematical facts:

1. The ring $\mathbb{Z}[\omega]$ has unique factorization, making it analogous to the Gaussian integers $\mathbb{Z}[i]$.
2. Eisenstein primes exhibit a *hexagonal* symmetry, as opposed to the square symmetry of Gaussian primes.
3. The modulo 3 and modulo 12 residue structure of rational primes determines their factorization behavior in $\mathbb{Z}[\omega]$.

A visualization of these concepts makes these abstract algebraic relationships immediately apparent through color and pattern.

Ulam versus Eisenstein-Ulam

Starting with a central number of $n = 41$, from Gauss's quadratic prime generator, $p(n) = n^2 + n + 41$, we notice prime alignment along principal directions. Algebraically, this polynomial is associated with the discriminant $D = 1^2 - 4(1)(41) = -163$. The fact that -163 is a Heegner number connects this generator to fundamental concepts in number theory, and the near-integer value of the Ramanujan constant, $e^{\pi\sqrt{163}} \approx 262537412640768743.999999999999925\dots$, reflecting the unique properties of the quadratic field $\mathbb{Q}(\sqrt{-163})$. This near-integer reflects the exceptional discriminant structure that makes the Euler polynomial so effective.

The classical **Ulam spiral** is constructed on a square lattice, inherently linked to the 4th roots of unity $(\pm 1, \pm i)$. This structure privileges four principal directions (horizontal, vertical, and two diagonals), and prime alignments algebraically tied to numbers expressible by quadratic forms.



Figure 1: The **Ulam spiral** with 41 at center, defined by the square lattice (roots of unity i), shows overwhelming prime density along the Euler generator's diagonal.

When 41 is placed at the center, the prime density is dominated by a single principal axis, the **SW-to-NE diagonal**. For $n = 0, 1, 2, \dots, 39$, $p(n)$ produces 40 consecutive primes, an exceptional streak that creates the dominant SW-NE diagonal visible in Figure 1. Ulam spiral's square geometry privileges certain quadratic progressions and $p(n)$ generates primes along a single diagonal, creating asymmetric density. The hexagonal Eisenstein-Ulam spiral, by contrast, distributes primes across six symmetric rays, providing balanced visualization of the mod 3 splitting behavior.

The **Eisenstein-Ulam spiral**, by contrast, is mapped onto a hexagonal lattice, whose geometry is defined by the 3rd roots of unity (1 and ω, ω^2). The hexagonal structure, with its six directional axes, allows for a visualization of primes related to the quadratic field $\mathbb{Q}(\sqrt{-3})$. When the center $N = 41$ is selected, the alignment of primes becomes dramatically resolved based on their residue modulo 3, reflecting their behavior in the Eisenstein integer ring $\mathbb{Z}[\omega]$.

Eisenstein Prime Hexagonal Spiral (Center: 41, N=2000)

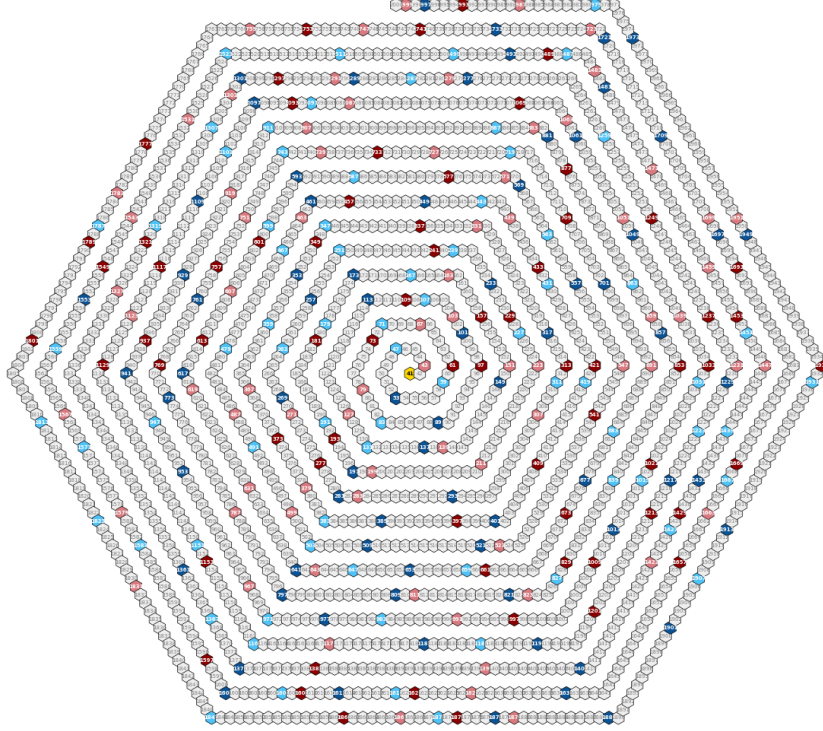


Figure 2: The **Eisenstein-Ulam** spiral with 41 at center uses the hexagonal lattice (roots of unity ω) revealing that $p \equiv 2 \pmod{3}$ primes (blue) cluster along NW-SW axes while $p \equiv 1 \pmod{3}$ primes (red/pink) concentrate along the E-W axis—a visual manifestation of splitting in $\mathbb{Z}[\omega]$

The hexagonal lattice elegantly separates primes based on their residue modulo 3:

1. **Eisenstein Primes** ($p \equiv 2 \pmod{3}$): These are the primes that remain inert (prime) in the Eisenstein integer ring $\mathbb{Z}[\omega]$, conspicuously clustering along the two directions that are **Northwest (NW)** and **Southwest (SW)** (in blue).
2. **Non-Eisenstein Primes** ($p \equiv 1 \pmod{3}$): These are the primes that split into two non-associate primes in $\mathbb{Z}[\omega]$. They cluster along the third principal direction, the **East-West axis** (in red and pink).

The interactive Claude driven [app](#) and [Gemini app](#) of the Eisenstein spiral allows one to explore how these primes segregate into distinct rays based on their residue modulo 12 and starting number. A snapshot is provided at the end of the document in Figure 8.

Structure of Primes in the Eisenstein Plane

The geometry of the Eisenstein integers becomes clearer when viewed not as abstract algebra but as a physical lattice, where arithmetic structure takes on visible spatial form.

Eisenstein Integers and Primes

The ring of Eisenstein integers is defined as:

$$\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\} = \left\{ a + b \cdot \frac{-1 + i\sqrt{3}}{2} : a, b \in \mathbb{Z} \right\}$$

An Eisenstein integer $\pi \in \mathbb{Z}[\omega]$ is an **Eisenstein prime** if it is non-unit and cannot be factored into non-units. The rational primes (primes in $\mathbb{Z}$) relate to Eisenstein primes as follows:

Theorem (Eisenstein Prime Characterization). *Let p be a rational prime. Then:*

- If $p = 3$, then p ramifies: $3 = -\omega^2(1 - \omega)^2$ where $1 - \omega$ is an Eisenstein prime.
- If $p \equiv 1 \pmod{3}$, then p splits: $p = \pi\bar{\pi}$ for distinct Eisenstein primes $\pi, \bar{\pi}$.
- If $p \equiv 2 \pmod{3}$, then p remains prime (inert) in $\mathbb{Z}[\omega]$.

The key insight is that primes $p \equiv 2 \pmod{3}$ are exactly the **Eisenstein primes** among rational primes. These are the primes we highlight in our visualization.

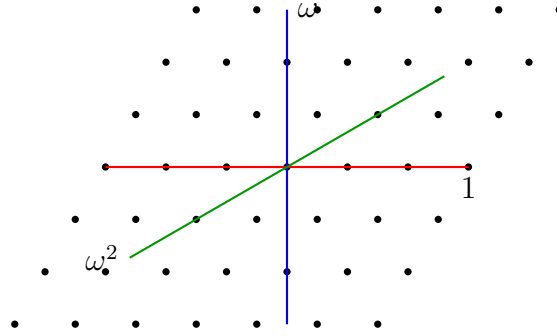


Figure 3: Triangular lattice structure of the Eisenstein integers $\mathbb{Z}[\omega]$. Points correspond to integers of the form $a + b\omega$.

Triangular lattice of $\mathbb{Z}[\omega]$, generated by 1 and $\omega = e^{2\pi i/3} = \frac{-1+i\sqrt{3}}{2}$. The coloured axes mark multiplication by the Eisenstein units $\{\pm 1, \pm\omega, \pm\omega^2\}$, forming the six principal rays. When integers are placed outward in the Eisenstein-Ulam spiral these become linear progressions $3n + c_k$, restricting primes on each ray to two residue classes modulo 12. The diagram expresses the sixfold symmetry governing prime behaviour in $\mathbb{Z}[\omega]$.

The Modulo 12 Classification

To understand the interaction between Eisenstein primes and the classical Fermat/Gaussian prime dichotomy, we refine the modulo 3 criterion using the Chinese Remainder

Theorem. A prime $p > 3$ satisfies both:

$$\begin{aligned} p &\equiv r_3 \pmod{3}, & r_3 &\in \{1, 2\} \\ p &\equiv r_4 \pmod{4}, & r_4 &\in \{1, 3\} \end{aligned}$$

By CRT, this gives four residue classes modulo 12:

$$\begin{aligned} p &\equiv 1 \pmod{12} &\Leftrightarrow & p \equiv 1 \pmod{3} \text{ and } p \equiv 1 \pmod{4} \\ p &\equiv 5 \pmod{12} &\Leftrightarrow & p \equiv 2 \pmod{3} \text{ and } p \equiv 1 \pmod{4} \\ p &\equiv 7 \pmod{12} &\Leftrightarrow & p \equiv 1 \pmod{3} \text{ and } p \equiv 3 \pmod{4} \\ p &\equiv 11 \pmod{12} &\Leftrightarrow & p \equiv 2 \pmod{3} \text{ and } p \equiv 3 \pmod{4} \end{aligned}$$

This yields our classification scheme:

$p \bmod 12$	$p \bmod 3$	$p \bmod 4$	Classification
1	1	1	Non-Eisenstein Fermat
5	2	1	Eisenstein Fermat
7	1	3	Non-Eisenstein Gaussian
11	2	3	Eisenstein Gaussian

Table 1: Prime classification by residue class modulo 12

Note that primes $p = 2$ and $p = 3$ are special cases: $2 \equiv 2 \pmod{12}$ but is even, while 3 ramifies in $\mathbb{Z}[\omega]$. The classification applies to all primes $p > 3$.

The two **Eisenstein prime classes** (in blue) are precisely those with $p \equiv 2 \pmod{3}$, which split further into:

- **Eisenstein-Fermat primes:** $p \equiv 5 \pmod{12}$ (also satisfy $p \equiv 1 \pmod{4}$)
- **Eisenstein-Gaussian primes:** $p \equiv 11 \pmod{12}$ (also satisfy $p \equiv 3 \pmod{4}$)

Combined Mod 4 and Mod 6 Classification

All primes greater than three satisfy the relation

$$p \equiv 1 \pmod{6} \quad \text{or} \quad p \equiv 5 \pmod{6}.$$

Meanwhile, the classical Fermat and Gaussian families distinguish primes by

$$p \equiv 1 \pmod{4} \quad \text{or} \quad p \equiv 3 \pmod{4}.$$

To classify primes simultaneously by their splitting behavior in $\mathbb{Z}[i]$ and $\mathbb{Z}[\omega]$, we combine both residues. The natural modulus is

$$\text{lcm}(4, 6) = 12.$$

Thus primes lie in four congruence classes:

$$p \equiv 1, 5, 7, 11 \pmod{12},$$

corresponding respectively to

- 1 : Non-Eisenstein Fermat
- 5 : **Eisenstein Fermat**
- 7 : Non-Eisenstein Gaussian
- 11 : **Eisenstein Gaussian**.

Visualization Design

The hexagonal lattice is constructed using axial coordinates (q, r) , where each hexagon is identified by two integer coordinates. The mapping to Cartesian coordinates is:

$$\begin{aligned}x &= \text{size} \cdot (q + r/2) \\ y &= \text{size} \cdot (r \cdot \sqrt{3}/2)\end{aligned}$$

The Ulam spiral walk follows six principal directions:

$$\text{directions} = [(1, 0), (0, 1), (-1, 1), (-1, 0), (0, -1), (1, -1)]$$

Starting from $(0, 0)$ containing the integer 1, we spiral outward in layers, placing consecutive integers at each hexagonal position. The color scheme directly reflects the modulo 12 structure:

- **Royal Blue (RGB: 65,105,225)**: Eisenstein-Fermat primes ($p \equiv 5 \pmod{12}$)
- **Sky Blue (RGB: 135,206,235)**: Eisenstein-Gaussian primes ($p \equiv 11 \pmod{12}$)
- **Dark Red (RGB: 139,0,0)**: Non-Eisenstein Fermat primes ($p \equiv 1 \pmod{12}$)
- **Light Pink (RGB: 255,182,193)**: Non-Eisenstein Gaussian primes ($p \equiv 7 \pmod{12}$)
- **Light Gray**: Composite numbers

The two blue shades (Eisenstein primes) are further distinguished by semi-transparent halos to emphasize their special status. All color encodings are supplemented by point size and halo presence, ensuring pattern distinguishability for colorblind readers.

Implementation Choices

Prime Testing Algorithm

We implement the standard trial-division primality algorithm using the $6k \pm 1$ optimisation:

test divisibility by 2 and 3, then by all integers of the form $6k \pm 1$ up to $\sqrt{n}$,

which yields a time complexity of $O(\sqrt{n})$ per test. This is sufficient for visualization ranges up to 10^4 – 10^5 , where full determinism and graphical clarity are more important than asymptotic speed. For larger-scale computation beyond $n = 10^6$, probabilistic algorithms such as Miller–Rabin offer significantly improved performance while retaining practical certainty.

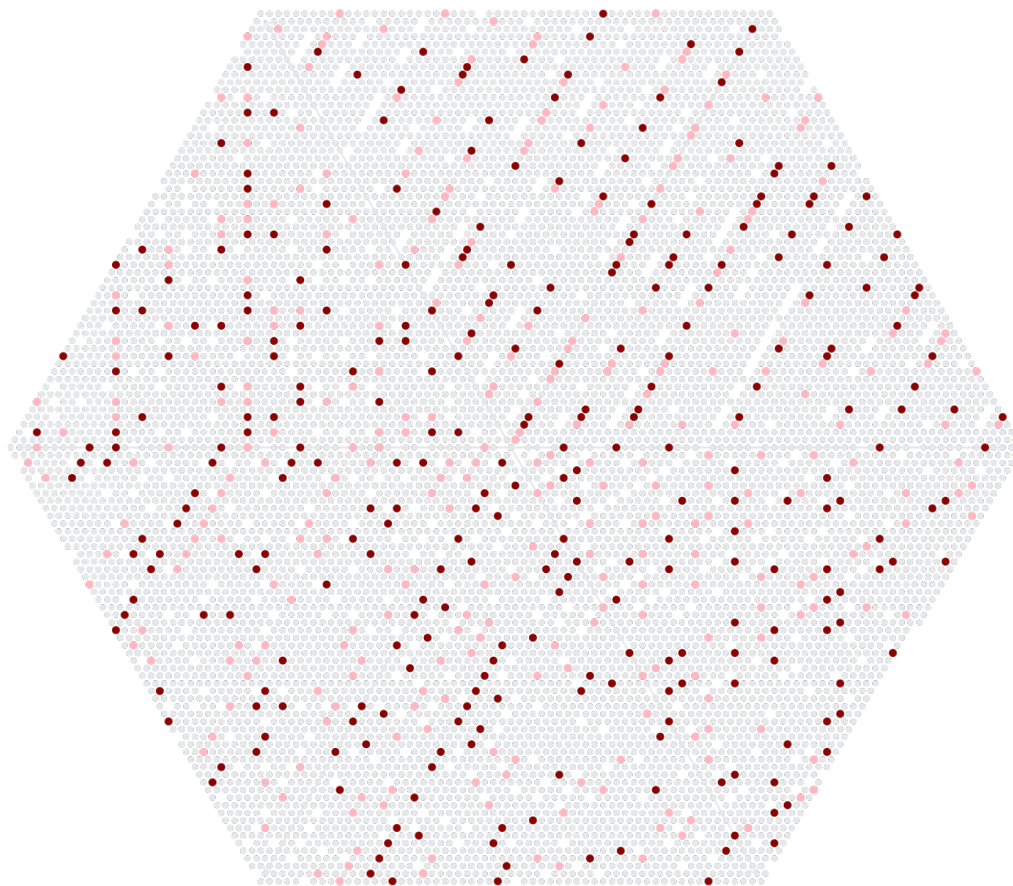
Canvas Rendering

We use HTML5 Canvas with 2D context for efficient rendering. Key optimizations:

- Base spacing scales with zoom: $\text{baseSize} = 4 \times \text{zoom}$
- Halos drawn first (z-index layering), followed by dots, then optional numbers
- Point sizes: Eisenstein primes ($2.2 \times \text{zoom}$), other primes ($1.8 \times \text{zoom}$), composites ($1.3 \times \text{zoom}$)

Visual Pattern Analysis

Non-Eisenstein primes



Non-Eisenstein Primes (Dark Red and Light Pink)

Range: 1–10000

Filters: Only Non-Eisenstein Primes enabled

Dark red (Fermat $4n+1$, $p \equiv 1 \pmod{12}$) and light pink (Gaussian $4n+3$, $p \equiv 7 \pmod{12}$) primes form diagonal patterns across the hexagonal lattice

Figure 4: Non-Eisenstein primes exhibit diagonal ray patterns reminiscent of Ulam's square spiral, but adapted to hexagonal symmetry. The dark red points show concentration along certain hexagonal rays, while light pink points fill complementary regions.

The composite numbers (gray) occupy the majority of lattice sites, yet their distribution is far from uniform. Large composite-free regions appear along certain rays, corresponding to twin prime sequences. The non-Eisenstein primes ($p \equiv 1 \pmod{3}$) split completely in $\mathbb{Z}[\omega]$. These primes follow patterns determined by quadratic residues modulo 3, creating visible diagonal structures. The dark red Fermat primes ($p \equiv 1 \pmod{12}$) cluster along specific rays, while the light pink Gaussian primes ($p \equiv 7 \pmod{12}$) show complementary distribution, reflecting the partition of primes $\equiv 1 \pmod{3}$ by their residue modulo 4.

Eisenstein primes

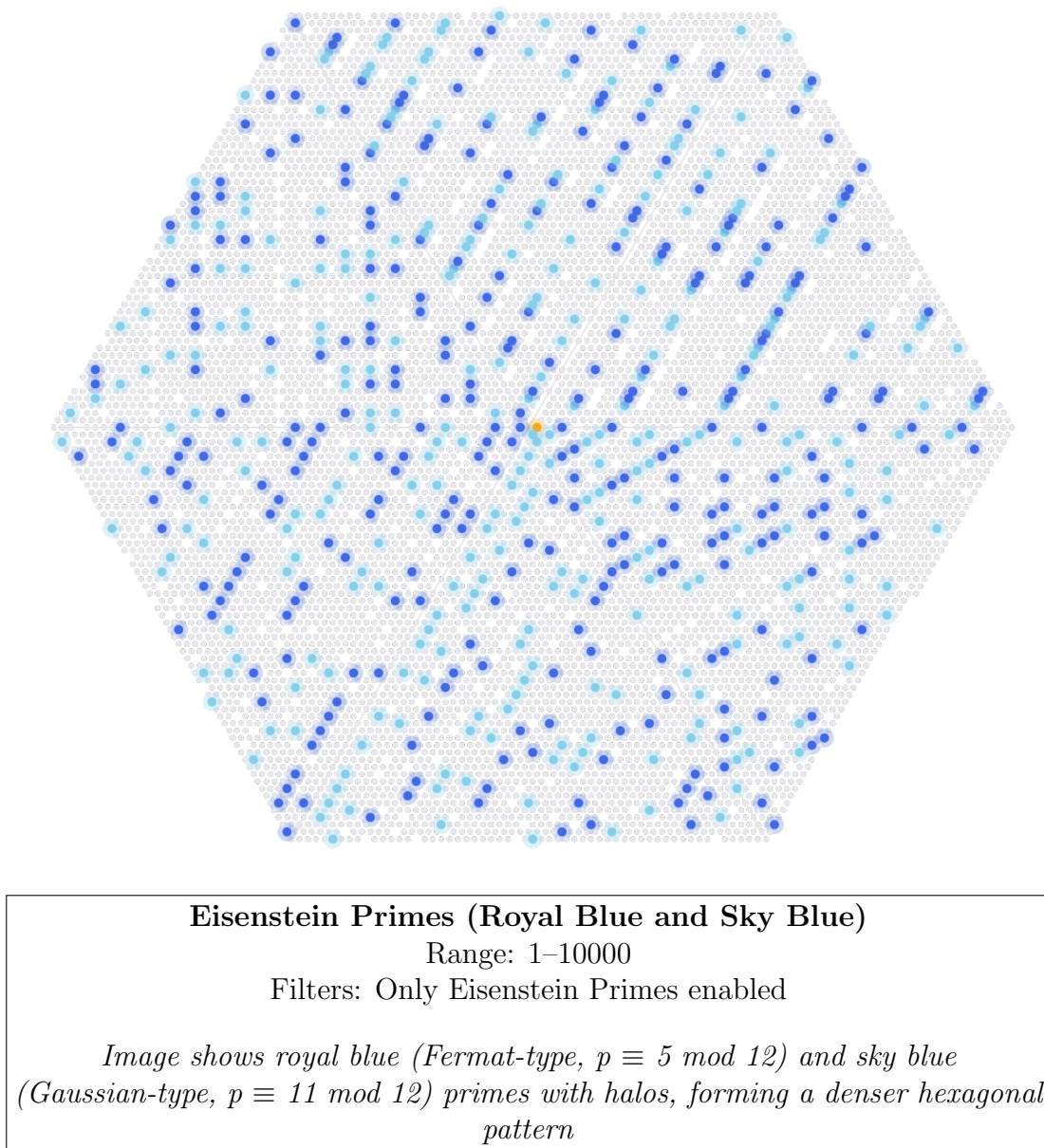


Figure 5: Eisenstein primes reveal a fundamentally different structure. The hexagonal symmetry is more pronounced, with both royal blue (Fermat-type) and sky blue (Gaussian-type) Eisenstein primes interwoven in a pattern reflecting the six-fold rotational symmetry of $\mathbb{Z}[\omega]$.

The Eisenstein primes ($p \equiv 2 \pmod{3}$) remain inert in $\mathbb{Z}[\omega]$ and show markedly different distribution patterns. The royal blue primes ($p \equiv 5 \pmod{12}$) are Fermat primes that happen to be Eisenstein primes, while sky blue primes ($p \equiv 11 \pmod{12}$) are Gaussian primes that are also Eisenstein primes. Their interleaved pattern reflects the fundamental hexagonal geometry of the Eisenstein integers, with both types contributing equally to the six principal rays emanating from the origin.

Hexagonal Ray Structure

Principal Rays of the Eisenstein Lattice

The hexagonal lattice $\mathbb{Z}[\omega]$ is generated by

$$1, \quad \omega = e^{2\pi i/3} = (-1 + i\sqrt{3})/2, \quad \omega^3 = 1, \quad \omega \neq 1.$$

The six principal rays correspond to multiplication by the units

$$\{\pm 1, \pm\omega, \pm\omega^2\}.$$

Each ray can be expressed as

$$\{n\omega^k : n \in \mathbb{Z}_{\geq 0}\}, \quad k = 0, \dots, 5,$$

and in the integer-labeled Eisenstein-Ulam spiral this becomes a linear arithmetic progression

$$p(n) = 3n + c_k.$$

Thus primes on each ray occupy at most two congruence classes modulo 12, explaining the visible diagonal filaments in the spiral.

The six rays below arise from multiplication by the units of $\mathbb{Z}[\omega]$, separated by 60° . In the Eisenstein-Ulam spiral these become linear sequences $3n + c_k$, producing the prime-bearing diagonal filaments observed in the visualization.

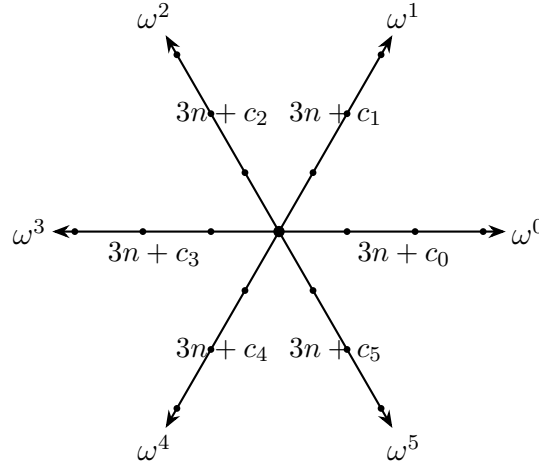


Figure 6: The six principal Eisenstein rays generated by powers of $\omega = e^{2\pi i/3}$. Each ray corresponds to an arithmetic progression $3n + c_k$ in the Eisenstein-Ulam spiral.

Eisenstein primes appear to concentrate along rays corresponding to these directions, suggesting a deep connection between:

- The arithmetic of $\mathbb{Z}[\omega]$
- Quadratic forms $x^2 + xy + y^2$ (norms in $\mathbb{Z}[\omega]$)
- The distribution of primes in arithmetic progressions modulo 3 and 12

Density Observations

By Dirichlet's theorem on primes in arithmetic progressions, the four residue classes modulo 12 occur with equal asymptotic density¹:

$$\#\{p \leq x : p \equiv a \pmod{12}\} \sim \frac{\pi(x)}{4}.$$

Eisenstein primes ($p \equiv 5, 11 \pmod{12}$) appear in the two blue tones, and collectively form roughly half of all primes. The remaining primes ($p \equiv 1, 7 \pmod{12}$), shown in red and pink, fill the complementary half. The sixfold symmetry of the hexagonal lattice causes the Eisenstein primes to align into distinct ray-like filaments, while the non-Eisenstein primes populate interstitial bands, giving a clear geometric expression to the analytic density predicted by Dirichlet.

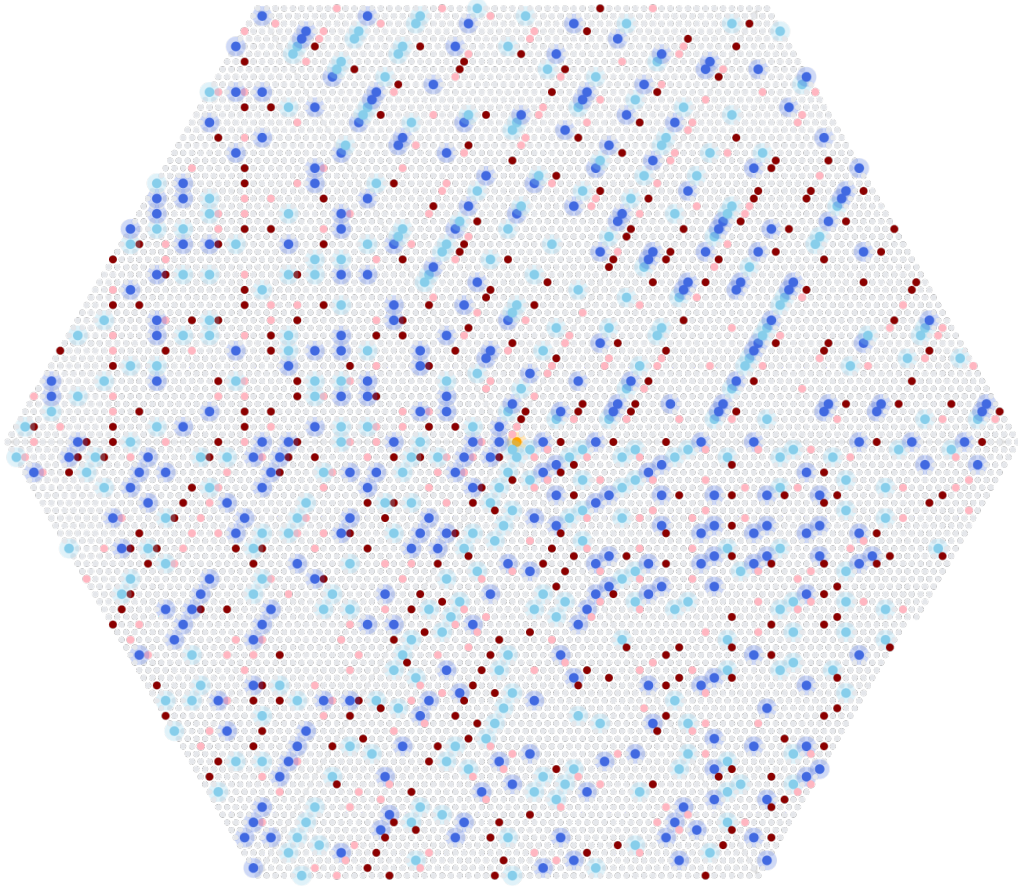


Figure 7: Visualization of primes on the Eisenstein-Ulam hexagonal spiral, range 1–10000.

Range 1–10000: Eisenstein primes ≈ 612 (50.1%), non-Eisenstein primes ≈ 608 (49.9%), total primes = 1229.

¹More precisely, by Dirichlet's theorem on primes in arithmetic progressions,

$$\pi(x; 12, a) \sim \frac{\pi(x)}{\varphi(12)} = \frac{\pi(x)}{4} \quad \text{as } x \rightarrow \infty,$$

where $\pi(x; 12, a)$ denotes the number of primes $p \leq x$ satisfying $p \equiv a \pmod{12}$.

Number Theoretic Significance

The Eisenstein-Ulam spiral gives geometric intuition for cubic reciprocity, the counterpart of quadratic reciprocity in the setting of the Eisenstein integers $\mathbb{Z}[\omega]$. Primes congruent to 2 (mod 3) remain inert and appear along the dominant blue filaments, while primes congruent to 1 (mod 3) split and occupy the complementary bands. What is typically expressed through algebraic rules on splitting and inertia becomes visible as symmetry and pattern.

The underlying modulo 12 classification reflects the decomposition law in the cyclotomic field $\mathbb{Q}(\omega)$, the simplest case of class field theory. The visual separation between Eisenstein and non-Eisenstein primes suggests computational strategies for primality testing, factorization, and cubic residue evaluation that exploit geometric regularity rather than randomness.

The spiral thus acts as a laboratory where algebraic number theory becomes visible, and where reciprocity appears not as an abstract theorem but as a geometric trace in the distribution of primes.

Future Directions

The Eisenstein-Ulam spiral opens a range of questions that extend beyond the visual patterns presented here. The emergence of structured rays and density symmetries hints that this approach may provide a broader framework for exploring number theoretic behaviours through geometric and computational methods.

We can generalize the Eisenstein-Ulam construction to spirals in $\mathbb{Z}[\zeta_n]$ for $n > 3$, where ζ_n is a primitive n th root of unity. This involves higher-order cyclotomic lattices (e.g. the 10-fold symmetry of $\mathbb{Z}[\zeta_5]$), which is implemented in the [app](#).



Figure 8: Visualization of primes on the Cyclotomic-5-Ulam spiral, range 41–15000.

The ray formation show five principal prime directions.

Several natural directions suggest themselves for further investigation:

1. **Higher-dimensional lattices and cyclotomic extensions:**

3D visualizations of $\mathbb{Z}[\omega]$ with a vertical axis encoding the norm $N(a + b\omega) = a^2 - ab + b^2$, and animated constructions showing the temporal emergence of prime filaments in higher symmetry settings.

2. **Norm surfaces:** 3D visualization of Eisenstein integer norms

3. **Statistical analysis:** Quantifying ray densities and angular distributions

4. **Dirichlet L -functions:** Connecting visual ray patterns to zeros of $L(s, \chi)$ where χ is the character defining splitting in $\mathbb{Z}[\omega]$, potentially revealing geometric signatures of the Generalized Riemann Hypothesis. [8]

Conclusion

The Eisenstein-Ulam spiral demonstrates that abstract algebraic structures in number theory can be made visually accessible through thoughtful design. By combining:

- The hexagonal geometry natural to $\mathbb{Z}[\omega]$;
- The modulo 12 classification reflecting both Eisenstein and Fermat/Gaussian structure;
- Interactive controls for exploration;

we create an intuitive window into the distribution of Eisenstein primes. The striking difference between Figures 3 and 4 makes immediately apparent what would otherwise require significant algebraic machinery to establish: that primes decompose into fundamentally different geometric families depending on their residue class modulo 3.

Interactive Visualization and Usage

The visualization tool for exploring prime distribution patterns serves as an educational resource for introducing students to algebraic number theory through visual intuition.

Interactive Application: <https://eisenstein-ulam.vercel.app>

Source Code: <https://github.com/leelemacjames/eisensteinUlam>

Cyclotomic Application: <https://cyclotomic-ulam.vercel.app/>

Source Code: <https://github.com/leelemacjames/cyclotomicUlam.git>

All code is licensed under MIT. Visualization parameters, rendering algorithms, and test cases are documented in the repository README.

Interactive Features

- **Range selection:** START and END parameters allow focusing on specific ranges
- **Zoom:** Continuous zoom from $0.3\times$ to $3\times$ for detail inspection. Pump the zoom in and out to maximize the visualization within the viewport. Images extractable by right click.

- **Filters:** Toggles for "Only Eisenstein Primes" and "Only Non-Eisenstein Primes"

Rendering performance: 10 000 points $\approx$ 100 ms (interactive), 100 000 points $\approx$ 2 s (responsive), 1 000 000 points $\approx$ 45 s (large-scale analysis). Performance measured on standard hardware (2024 Lenovo Thinkpad).

Key Visual Findings

1. Eisenstein primes form 6 dominant radial filaments
2. Non-Eisenstein primes occupy complementary bands
3. Filament density decreases as $O(1/\log x)$ by the Prime Number Theorem
4. Angular uniformity confirms Dirichlet's equipartition

References

- [1] S. Ulam, *Patterns of growth of figures formed by random points*, in *Proceedings of Symposia in Applied Mathematics*, vol. 14, 1964, pp. 215–224.
- [2] M. Gardner, *Mathematical Games: The remarkable patterns hidden in numbers*, *Scientific American*, March 1964.
- [3] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.
- [4] D. A. Cox, *Primes of the Form $x^2 + ny^2$* , Wiley, 1989.
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- [6] J. Conway and R. Guy, *The Book of Numbers*, Springer, 2013.
- [7] L. Washington, *Introduction to Cyclotomic Fields*, 2nd ed., Springer GTM 83, 1997.
- [8] H. Davenport, *Multiplicative Number Theory*, 3rd ed., revised by H. L. Montgomery, Graduate Texts in Mathematics 74, Springer, 2000.

Spontaneous Symmetry Formation

Figure 9: Full Eisenstein-Ulam spiral visualization, range 41–15000, showing all four prime classes simultaneously. The hexagonal symmetry and radial filament structure are clearly visible. Blue tones (Eisenstein primes) dominate NW and SW rays while red/pink tones (non-Eisenstein primes) occupy Easterly ray.

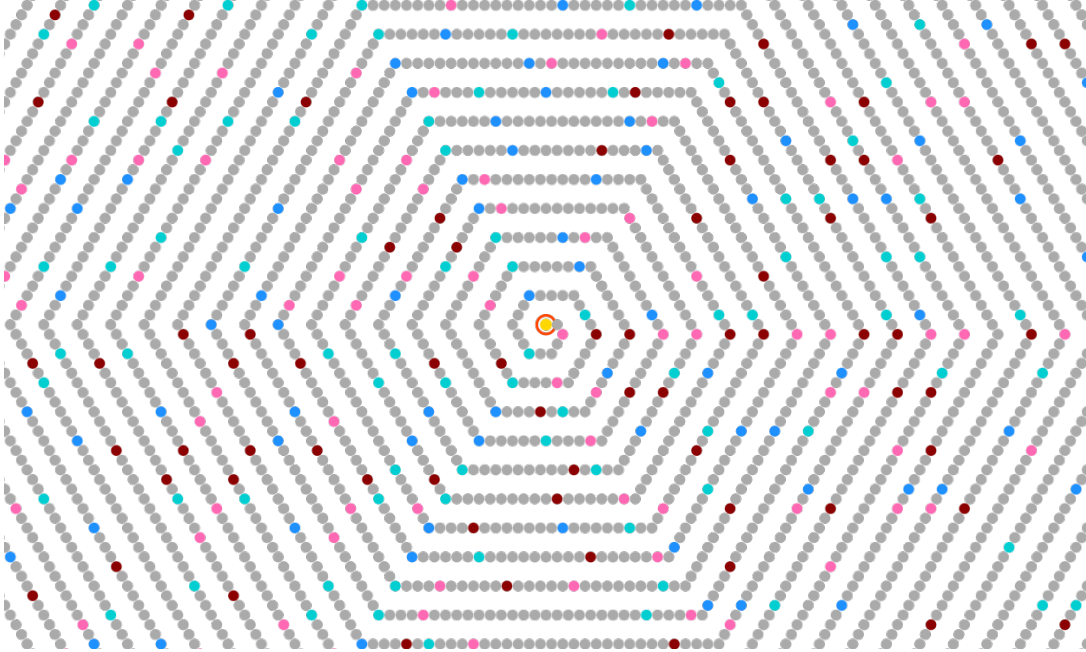


Figure 9: Visualization of primes on the Eisenstein-Ulam hexagonal spiral, range 41–15000.

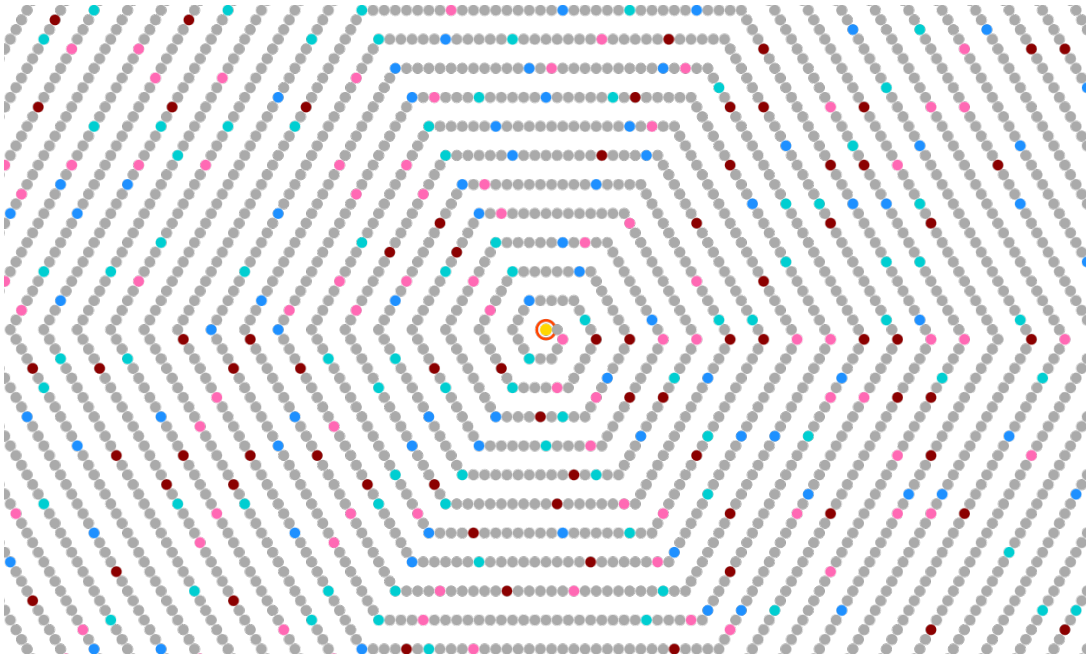


Figure 10: Visualization of primes on the Eisenstein-Ulam hexagonal spiral, range 67–15000.

The same ray formation returns with the second largest Heegner number of 67.